Convert fractions to decimals

Use the scaffold, show each step and ask for an adult check when marked.

Name: ______________________________ Date: ________________

1. Read the model: Write 7/20 as a decimal. Copy or complete the first step.

2. Write 0.625 as a percentage.

3. Write 35% as a fraction in simplest form.

4. Miro says: Miro moves the decimal point without considering the value. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Write 7/20 as a decimal. Copy or complete the first step.
   Answer 0.35.

2. Write 0.625 as a percentage.
   Answer 62.5%.

3. Write 35% as a fraction in simplest form.
   Answer 7/20.

4. Miro says: Miro moves the decimal point without considering the value. Is this correct? Explain.
   Answer Use a hundred-square model or multiplication and division by 100, then check against a known benchmark.

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Work independently. Show a complete method and check at least one answer.

Name: ____ Date: __________________

1. Write 7/20 as a decimal.

2. Write 0.625 as a percentage.

3. Write 35% as a fraction in simplest form.

4. Solve this independently and show every stage: Write 7/20 as a decimal.

5. Use a second method or representation for: Write 0.625 as a percentage.

6. Check this result using an inverse, estimate or known fact: Write 35% as a fraction in simplest form.

7. Prove or explain your reasoning: Order 3/5, 0.58 and 62% from least to greatest.

8. Identify and correct this misconception: Miro moves the decimal point without considering the value.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: Write 35% as a fraction in simplest form.

Teacher answers and checking prompts

1. Write 7/20 as a decimal.
Answer 0.35.

2. Write 0.625 as a percentage.
Answer 62.5%.

3. Write 35% as a fraction in simplest form.
Answer 7/20.

4. Solve this independently and show every stage: Write 7/20 as a decimal.
Answer 0.35.

5. Use a second method or representation for: Write 0.625 as a percentage.
Answer Expected result: 62.5%. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: Write 35% as a fraction in simplest form.
Answer Expected result: 7/20. A valid check must be shown.

7. Prove or explain your reasoning: Order 3/5, 0.58 and 62% from least to greatest.
Answer 0.58, 3/5, 62%. The explanation must show why the method works.

8. Identify and correct this misconception: Miro moves the decimal point without considering the value.
Answer Use a hundred-square model or multiplication and division by 100, then check against a known benchmark.

9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: Write 35% as a fraction in simplest form.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

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Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ______________________________ Date: ____________

1. Prove or explain your reasoning: Order 3/5, 0.58 and 62% from least to greatest.

2. Prove the answer to this question, rather than only calculating it: Write 7/20 as a decimal.

3. Solve this in two different ways and compare them: Write 0.625 as a percentage.

4. Work backwards from the answer to reconstruct the question: 7/20.

5. Explain why this reasoning fails: Miro moves the decimal point without considering the value.

6. Create a new example that tests the same idea as: Write 35% as a fraction in simplest form.

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Order 3/5, 0.58 and 62% from least to greatest.
   Answer 0.58, 3/5, 62%. The explanation must show why the method works.

2. Prove the answer to this question, rather than only calculating it: Write 7/20 as a decimal.
   Answer Expected result: 0.35. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: Write 0.625 as a percentage.
   Answer Expected result: 62.5%. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 7/20.
   Answer One valid reconstruction is: Write 35% as a fraction in simplest form. Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro moves the decimal point without considering the value.
   Answer Use a hundred-square model or multiplication and division by 100, then check against a known benchmark.

6. Create a new example that tests the same idea as: Write 35% as a fraction in simplest form.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.